

Single-Round Vector RAG vs an LLM-Compiled Wiki: A Preregistered Comparison on a Small Multi-Domain Research Corpus

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Abstract

We preregistered a comparison of two ways to help an LLM answer questions over a small research corpus: a single-round Vector RAG system and an LLM-compiled markdown wiki browsed by a tool-using agent. Both systems answered the same 13 questions over 24 papers using the same answer-generating model, and their answers were scored by two blinded LLM judges.

The three preregistered predictions, reported here in their registered order, came out one weakly supported, one supported, and one refuted. First, the wiki was predicted to synthesize better across papers, on both cross-paper connection and answer organization. It scored much better at connecting findings, but its organization advantage fell below the registered threshold once both judges' scores were combined, leaving this prediction only weakly supported. Second, RAG was predicted to hold its own on single-fact lookup questions, and it met the registered test on the scored groundedness criterion, though that result was judge-sensitive: the second judge alone would have refuted it, and the preregistered Bayesian check did not corroborate it. Third, the wiki was predicted to be expensive to build and cheap to query, paying for itself after enough questions. The build side held, by roughly two orders of magnitude, but the query side reversed: the wiki spent about 21 times more LLM tokens per query than RAG, so the cheaper queries never arrive and no break-even point exists. That prediction is refuted.

With two exploratory analyses, the results tell a four-part story about why such comparisons disagree. Against a single-round retrieval baseline, compilation looks strongly beneficial for multi-paper synthesis. A decomposition-retrieval variant of RAG then removes almost all of that advantage at lower token cost, showing the advantage is sensitive to retrieval coverage. Decomposition does *not* remove the wiki's advantage in claim-by-claim citation support, leaving a separate evidence-artifact alignment gap. And holistic groundedness scoring disagrees with atomized citation checking by direction, and also between judges: on that criterion the two judges' rank agreement is near zero ($\rho = 0.04$), against $\rho = 0.81$ on the most concretely defined criterion.

The conclusion is that grounded research synthesis is not a single capability: systems differ in how well they organize evidence, how well their citations support each claim, and what they cost to run, and no architecture here was best on all three. Which one appears to win depends on the retrieval baseline, the scoring granularity, and the judge.

1 Introduction

An "LLM Wiki" turns a research corpus into a cross-linked wiki at ingest time, and at question time the model browses that wiki rather than retrieving raw chunks from the original papers. This architecture was popularized in informal commentary, notably Andrej Karpathy's "agentic

markdown wiki” framing. The idea suggests three possible tradeoffs: the wiki may help with multi-paper synthesis; rewriting papers into wiki pages may lose source fidelity; and the wiki may move cost from query time to ingest time, with a break-even point after enough questions. Compiled-knowledge architectures have since been compared against retrieval baselines by several groups, with materially different results depending on how the compiled knowledge is served and what is measured (Section 2). We do not claim priority; we claim a design, and the contribution of this paper is what a preregistered, blinded, two-judge protocol with claim-level citation scoring shows about *why* such comparisons disagree.

This paper reports a preregistered, blinded, two-judge comparison of Vector RAG and LLM Wiki on a fixed 24-paper corpus answering 13 evaluation questions across six difficulty tiers. Both systems use the same query model (Claude Opus 4.7 at **xhigh**); the cross-family judge is GPT-5.4 with Gemini 2.5 Pro for inter-rater reliability. Corpus, question set, rubric, decision rules, and Bayesian model were locked at the OSF preregistration tag before any judge run.

Contributions.

- **Preregistered comparison.** The wiki was much better at connecting papers; its advantage in answer organization was weaker after judge adjustment, positive but below the registered $H_1 + 2.0$ threshold (the rubric criteria are `inter_paper_mapping` and `structural_integrity`). RAG met the preregistered test for single-fact source grounding on `groundedness` (small-effect-fragile under IRR adjustment). The expected wiki cost advantage failed in the opposite direction: the wiki spent about $21\times$ more per query than RAG.
- **Evaluation-method finding.** Inter-judge agreement varied sharply across rubric criteria, and the spread is extreme: it was highest on the most operationally concrete criterion and lowest on the most holistic one, Spearman $\rho = 0.81$ (ICC 0.81) on `inter_paper_mapping`, the one criterion with a countable definition, against $\rho = 0.04$ (ICC 0.04) on holistic `groundedness`, where Gemini 2.5 Pro saturates near 10/10 and GPT-5.4 uses more of the scale. A post-hoc claim-level grounding analysis disagrees with the rubric `groundedness` score by direction, indicating that holistic and per-citation grounding metrics measure related but distinct properties. This suggests criterion design, not judge identity alone, can materially affect LLM-judge reliability; agreement should be measured per criterion rather than assumed uniform across a rubric.
- **Exploratory RAG ablation (two-judge).** A decomposition-retrieval variant of RAG recovered nearly all of the wiki’s advantage on cross-paper synthesis, closing $\sim 94\%$ of the gap on the H_1 subset under the same IRR adjustment the confirmatory analysis uses, or $\sim 88\%$ scored by the primary judge alone, and leaving a remaining wiki advantage smaller than what the registered $+2.0$ threshold would have required. It did *not* recover the wiki advantage in claim-by-claim citation support. The result is a three-way tradeoff among single-round RAG, decomp-RAG, and wiki along synthesis structure, claim-citation alignment, and cost.

One deviation from the registered procedure. The registration states that the RAG run, the wiki-side run, and the judge would not be invoked before registration acceptance. The RAG run and the judge complied; the **wiki-side run was invoked 77 minutes before the registration timestamp**. We report this here rather than leave it to be found: Section 3 gives the full chronology, what it does and does not put at risk, and the frozen-instrument timestamps that bound it.

The five preregistered run artifacts and five post-hoc artifacts are deposited on OSF with SHA-256 hashes; the reproducer scripts ship in the supplementary material.

2 Related Work

RAG and multi-hop retrieval. Retrieval-augmented language modeling was introduced by REALM [Guu et al., 2020] and Lewis et al. [Lewis et al., 2020] with dense passage retrieval [Karpukhin et al., 2020]. Our Vector RAG is a single-round retrieve-then-generate instantiation with multi-query expansion, hybrid retrieval, Cohere reranking, and CRAG-inspired [Yan et al., 2024] corrective validation; it does not perform retrieval-during-generation (cf. IRCOT [Trivedi et al., 2023], FLARE [Jiang et al., 2023]). Multi-hop benchmarks (HotpotQA [Yang et al., 2018], MuSiQue [Trivedi et al., 2022], MultiHop-RAG [Tang and Yang, 2024]) document that single-round retrieval is insufficient for connected reasoning across documents.

Hierarchical and abstractive memory. The LLM Wiki architecture sits closest to compiled abstractive memory: an offline LLM-built representation queried in lieu of (or alongside) raw chunks. RAPTOR [Sarathi et al., 2024] builds a recursive cluster-summary tree; RECOMP [Xu et al., 2024] compresses retrieved documents; GraphRAG [Edge et al., 2024] builds an entity graph with community summaries; knowledge-graph variants include REANO [Fang et al., 2024] and GNN-RAG [Mavromatis and Karypis, 2025]. Our wiki is conceptually an LLM-built version of this family with a markdown-rendered intermediate representation. OpenScholar [Asai et al., 2026] and PaperQA2 [Skarlinski et al., 2024] apply similar primitives at much greater scale.

Contemporaneous compiled-knowledge comparisons, and why they disagree. Three studies bear directly on the present comparison, and their results do not line up with each other or with ours. Huerta [2026] compiles a wiki memory and serves it as full context via KV caching over 17 RepLiQA domains and 6,800 questions, reporting that full-context inference beats RAG on curated knowledge (4.38 vs 4.08 of 5) but degrades below RAG at larger scale through attention dilution, and that blind compilation loses substantial quality before iterative refinement recovers it. Zhang [2026] compares an offline-compiled wiki against BM25 RAG on 28 cross-institutional policy queries with a shared generator, and reports the wiki ahead on citation recall (0.242 vs 0.095) while consuming a *fraction* of the retrieval baseline’s query-time context. Chan et al. [2024] argues more broadly that for knowledge bases small enough to fit in context, preloading and caching removes the retrieval step altogether. Our per-query result runs opposite to Zhang’s. The results are not directly contradictory, because the studies evaluate substantially different operating points, differing in corpus, task, generator, retrieval baseline, scale, and metrics as well as in architecture. Serving strategy is an especially salient difference: their wiki is *served* as retrieved pages, and Chan et al.’s and Huerta’s from a cached context window, whereas ours is browsed by a tool-using agent that pays for its own traversal, up to 30 turns of it. We cannot isolate that factor across studies. This is the paper’s central point in miniature. “Wiki” and “RAG” name families of systems, not operating points, and a comparison can reverse its cost verdict under a different serving strategy. It is also why we report synthesis structure, claim-citation alignment, and cost as three separate axes rather than one.

Citation-faithfulness and LLM-as-judge. ALCE [Gao et al., 2023] introduced citation-precision/recall benchmarks for long-form answers; Liu et al. [2023a] found commercial generative search systems often produce fluent but unsupported answers; our claim-level grounding analysis follows this protocol family. Zheng et al. [2023] document position, verbosity, and self-enhancement biases in LLM judges; G-Eval [Liu et al., 2023b] reports improved human alignment under structured form-filling but notes possible bias toward LLM-generated text; Wang et al. [2024] show ranking

can change with response order. Our IRR ceiling-judge finding (Section 5.3) is consistent with this literature.

3 Methods

Corpus and questions. 24 peer-reviewed papers across three domains (8 each: AI ethics & law, climate science, precision medicine), 2017–2026. 13 evaluation questions across six tiers: chronological, conflict, multi-hop, emergence, policy ($n = 2$ each, preregistered as wiki-favoring), and bias-check ($n = 3$, preregistered as RAG-favoring with extra power for H_2). Inclusion: peer-reviewed or arXiv-preprint research papers; exclusion: review-only summaries, books, non-research artifacts. Corpus list with DOIs and full question text in the OSF deposit.

Vector RAG. RAG retrieves passages once, reranks and validates them, then writes the answer. Implementation: header-aware chunking (target 512 tokens, 50-token overlap), per-chunk contextual enrichment, multi-query expansion, dense+sparse hybrid retrieval, Cohere reranking to top-5, and CRAG-inspired [Yan et al., 2024] corrective validation. Exact model versions, prompts, and parameters for every stage are in the supplementary material. “Single-round” means no retrieval-during-generation (cf. [Trivedi et al., 2023, Jiang et al., 2023]); the pipeline is otherwise multi-stage.

LLM Wiki. The wiki system first compiles the papers offline into a persistent cross-linked markdown wiki (entities, concepts, sources, and analyses with cross-references). At question time, a tool-using agent lists pages, reads selected pages, and submits an answer via three tools (`list_pages`, `read_page`, `submit_answer`; `MAX_TURNS=30`, no query-side prompt caching). Two minor telemetry caveats ($\leq 7\%$ thinking-token over-count; $\leq 6.5\%$ upper-bound caching saving forgone on the query side) do not materially affect the per-query comparison.

Both systems use the same query model (Claude Opus 4.7 at `xhigh`), corpus, and evaluation set; the comparison is fully crossed and within-question paired. This is a comparison of two practical architectures, not a clean causal isolation of “knowledge organization”: the systems differ along multiple axes (offline LLM rewriting; compiled markdown vs vector store; tool-loop vs single-round retrieval; multi-call vs single-call context).

Judging. Primary: GPT-5.4 at `medium` reasoning, cross-family from the answer model. IRR: Gemini 2.5 Pro, same prompt, fixed seed offset. Per-question $\{\text{RAG, Wiki}\} \rightarrow \{A, B\}$ blinding via `random.Random(seed=42)` (primary) / `(seed=43)` (secondary). The judge prompt instructs the model to score each system independently on each criterion. Rubric: four 1–10 anchored criteria with explicit anchors at 1/5/10: `groundedness` (claims trace to source material), `structural_integrity` (unified narrative vs concatenated fragments), `conflict_awareness` (naming of contradictory findings), `inter_paper_mapping` (multi-hop synthesis across ≥ 2 papers).

Preregistration and decision rules (paraphrased).

- H_1 (synthesis): $\text{Wiki} - \text{RAG} \geq +2.0$ on `inter_paper_mapping` and `structural_integrity` for the multi-hop+emergence tier ($n = 4$). Supported if both clear; weakly supported if both positive but at least one below; refuted if either negative.
- H_2 (point-source): $\text{RAG} - \text{Wiki} \geq 0$ on `groundedness` for the bias-check tier ($n = 3$).
- H_3 (cost): both $T_{\text{ingest}}[\text{wiki}] > T_{\text{ingest}}[\text{rag}]$ AND $T_{\text{query}}[\text{rag}] > T_{\text{query}}[\text{wiki}]$.

The full preregistration is available at <https://doi.org/10.17605/OSF.IO/ZEMHP>. The preregistration is explicit that the primary analyses are directional comparisons of means against magnitude thresholds, not frequentist tests, because n is small and LLM scores have non-i.i.d. error structure. Two robustness checks are also preregistered (bootstrap percentile CI; Bayesian model with $\mu \sim \mathcal{N}(0, 4)$, $\sigma \sim \text{HalfNormal}(4)$, NUTS).

Chronology, and one deviation from the registered starting rule. Because the tier design labels five question tiers wiki-favoring and one RAG-favoring, the order of events matters to how much the registration constrains. All times UTC on 2026-05-06 unless noted.

Table 1: Execution chronology. Times are the artifacts’ own `started_at` fields, which each runner captures at invocation, and git author timestamps for the commits. Every row is checkable against the deposited artifacts; the supplementary material gives the commit identifiers.

Time (UTC)	Event	Evidence
05-05 01:41	Questions and rubric frozen	last commit to <code>questions/rubric.yaml</code>
20:08	Design freeze	the commit carrying the prereg tag
20:29	RAG-side ingest invoked	<code>ingest-...T202904Z</code>
20:55	Wiki-side run invoked	<code>wiki-run-...T205500Z</code>
22:12	OSF registration	<code>date_registered</code> , reg. <code>zemhp</code>
22:16	RAG-side run invoked	<code>run-...T221602Z</code>
22:56	Judge invoked	<code>judged-...T225645Z</code>

The registration’s starting rule states that “the 13-question RAG run, the Wiki-side run, and the judge will not be invoked before registration acceptance,” and separately discloses that RAG-side ingestion was already running. **The wiki-side run does not satisfy that rule:** it was invoked 77 minutes before the registration timestamp and completed 13 minutes after it. The RAG run and the judge do satisfy it, at +3 and +44 minutes. We report this rather than leave it to be discovered, and we have not restated the rule to fit what happened.

The deviation weakens the procedural preregistration claim, and we state the weakening rather than argue around it: the wiki run had begun before the OSF registration became immutable, and incidental exposure to its output cannot be excluded. Wiki answer text is outcome information even though no judge score existed yet.

Three facts bound the scope for outcome-contingent adaptation, and each is externally checkable. The question set and rubric were committed 43 hours earlier and unmodified after. The registered decision rules and thresholds are in the same frozen commit, which the registration cites by tag. And no judge score of any kind existed until 44 minutes *after* registration, so the registered analysis plan was fixed before any score was observable. These timestamps substantially constrain, but do not eliminate, the scope for the design to have followed the data.

IRR-triggered adjustment. If the maximum primary–secondary score delta exceeds 2 on any criterion, that criterion’s `mean_tier` values are recomputed using the mean of the two judges’ per-question scores in place of the primary-judge-only mean for the H_1/H_2 tests.

4 Experimental setup

Artifacts. Five preregistered run artifacts plus five post-hoc artifacts (claim-level grounding on the original run; the decomp ablation run, its judge run, and its grounding; and the two-judge

rejudge of the ablation) are deposited on OSF at <https://osf.io/j37b8/> with SHA-256 hashes. Every reproducer script cited in this paper, including the analysis that regenerates the reported values and the agreement statistics in Table 3, ships in the supplementary material accompanying this submission, which is self-contained and requires no repository access. *Manipulation check.* All 26 (question, system) cells used `claude-opus-4-7` at `xhigh`; zero exclusions, zero ingest/query/judge failures.

Cost-accounting scope. H₃ follows the preregistered token-accounting (sum of prompt + completion at ingest; prompt + completion + thinking at query) and excludes embedding spend, reranker spend, vector-store storage, and dollar normalization. *Terminology.* Because of those exclusions, and because the recovered ingest figure weights the input side for caching while leaving output tokens at face value, the quantity this paper measures is **LLM-token usage**, a token-accounting proxy for cost, and not billable cost. We write “cost” for it only where the token basis is unambiguous from context; ratios such as the 21× per-query figure are ratios of tokens, and they would not survive unchanged into dollars under provider pricing that bills token classes at different rates. *Prompt-cache caveat (load-bearing).* The wiki ingest artifact rolls `uncached` + `cache_creation` + `cache_read` into one field. Anthropic bills cache reads at ~10% of base input rate, so the gross figure overstates billable cost by an order of magnitude on cache-heavy workloads. The H_{3b} test is captured uncached on both sides and needs no correction. H_{3a} is adjudicated on the registered token metric directly from the deposited artifacts; because that metric’s economic weight is uneven across token classes, Section 5 additionally reports a post-hoc billing-weighted sensitivity analysis from per-session components that reconcile exactly to the deposited aggregate, as a range over the two possible cache-write rates.

Robustness checks (preregistered). *Bootstrap:* 95% percentile CI, 10,000 resamples per criterion × tier. *Bayesian:* per criterion × tier, `score_diff` $\sim \mathcal{N}(\mu, \sigma^2)$ with $\mu \sim \mathcal{N}(0, 4)$, $\sigma \sim \text{HalfNormal}(4)$, NUTS (4 chains × 2,000 iter × 2,000 warmup, `random_seed=42`); all fits met the preregistered convergence criteria ($\hat{R} \leq 1.003$, min ESS $\geq 1,907$). The Bayesian “strongly corroborated” threshold is $P(\mu \geq \theta) \geq 0.95$ ($\theta = +2$ for H₁, $\theta = 0$ for H₂).

IRR adjustment. The preregistration specified a 3-question IRR subset; during execution we extended secondary judging to all 13 questions, a disclosed rigor-increasing deviation. The max-delta trigger fires on all four criteria under both the prereg $n = 3$ and the expanded $n = 13$ coverage; the adjustment rule is invariant to the deviation.

Post-hoc claim-level grounding. For each of the 26 answer cells, Claude Opus 4.7 (medium adaptive thinking) atomizes the answer into atomic claims with `cited_source_idx`; for each cited claim, GPT-5.4 (medium, cross-family) scores against the cited chunks as `supported` / `partial` / `contradicted` / `unsupported`. Uncited claims auto-receive `unsupported`. Full prompts in the OSF supplement.

5 Results

We report each hypothesis under the **primary-only** reading (the pre-IRR baseline) and the **IRR-adjusted** reading (judge-mean recomputation; see Section 3). The trigger and adjustment rule were both preregistered. The IRR-adjusted reading is the more conservative of the two and is the basis for our headline verdicts.

5.1 Confirmatory analyses

H₁: Synthesis advantage ($n = 4$). Per-question paired diffs (Wiki – RAG) on the multi-hop + emergence subset are given in full in the supplementary material; the subset means are `structural_integrity` +2.000 primary / +1.250 secondary / +1.625 judge-averaged, and `inter_paper_mapping` +6.250 / +7.000 / +6.625.

Primary-only: $\Delta_{\text{struct}} = +2.000$ (lands exactly at threshold), $\Delta_{\text{mapping}} = +6.250$ (clears). Both criteria $\geq +2.0 \Rightarrow$ *supported*. *IRR-adjusted (headline)*: $\Delta_{\text{struct}} = +1.625$ (misses by 0.375), $\Delta_{\text{mapping}} = +6.625$ (clears) $\Rightarrow$ *weakly supported* per the preregistered three-way decision rule. The `inter_paper_mapping` advantage is large and never close to the threshold; the `structural_integrity` advantage is threshold-sensitive.

With $n = 4$ the most legible robustness check is to drop each question in turn. On `inter_paper_mapping` the judge-averaged mean stays between +5.833 and +7.833, never approaching +2.0, so that verdict does not depend on any single question. On `structural_integrity` it moves between +1.000 and +2.000: dropping `T3-rwd-validity-for-side-effects` lands exactly on the threshold and would flip the verdict from weakly supported to supported, and dropping `T4-eelgrass-mrv-gaps` halves the effect. The H₁ verdict is therefore two separate findings with very different evidential weight, and we treat the leave-one-out spread, not the third decimal place of the posterior, as the honest summary of what $n = 4$ supports.

H₂: Point-source parity ($n = 3$). Per-question paired diffs (RAG – Wiki) on `groundedness` for the bias-check tier, primary / secondary: B1 +1/+3, B2 +3/–6, B3 +2/+1. Means are +2.000 primary, –0.667 secondary, +0.667 judge-averaged.

The B2 secondary entry is the largest single judge disagreement in the experiment: GPT-5.4 scored RAG=9 / wiki=6, Gemini 2.5 Pro scored RAG=4 / wiki=10. Both readings nonetheless return $\Delta \geq 0$, satisfying the preregistered decision rule. *Primary-only*: $\Delta = +2.000 \Rightarrow$ *supported*. *IRR-adjusted*: $\Delta = +0.667 \Rightarrow$ *supported*.

We report this verdict as registered and do not relabel it after the fact, but the evidence behind it is thin and we state its shape rather than only its outcome. H₂ *met the preregistered point-estimate criterion, and was judge-fragile and not strongly corroborated by the preregistered Bayesian check*: the judge-averaged effect is +0.667 on a 1–10 scale, the secondary judge alone returns –0.667 and would refute, the Bayesian posterior gives $P(\mu \geq 0) = 0.660$ against the registered corroboration bar of 0.95, and the criterion it rests on is the one with near-zero inter-judge rank agreement (Table 3). Leave-one-question-out over the $n = 3$ subset keeps the sign but reaches the threshold exactly: dropping B1 gives $\Delta = +0.000$, B3 gives +0.250, B2 gives +1.750. Section 5.4 reports a post-hoc analysis that nuances the underlying mechanism.

H₃: Cost asymmetry. The preregistration expected the wiki to be expensive to build but cheap to query. The clean query-side data showed the opposite: the wiki cost more per query, so the amortization story cannot hold. Both sides are adjudicable directly from the registered token accounting. We then report a post-hoc economic sensitivity analysis, because the wiki’s registered ingest count is dominated by cache reads that a provider bills at a fraction of the base rate, and we want the verdict to be shown robust to that rather than resting on a count whose economic meaning is uneven.

The wiki query-side harness explicitly disabled prompt caching (companion §4.3), and RAG per-query is similarly uncached, so the per-query figures are directly observed uncached token counts and require no cache-class correction. The 21× per-query gap is a *token* ratio, not a dollar-cost ratio.

Quantity	Value
$T_{\text{query}}[\text{rag}]$, 13 questions (prompt + completion + thinking)	78,093
$T_{\text{query}}[\text{wiki}]$, 13 questions (prompt + completion + thinking)	1,651,357
$T_{\text{ingest}}[\text{rag}]$, 24 papers (prompt + completion)	1,401,996
$T_{\text{ingest}}[\text{wiki}]$, registered metric (tokens_in + tokens_out)	904,884,660
<i>post-hoc</i> billing-weighted input, 5-min / 1-hour TTL	135,702,160 / 161,494,660
H3a supported ($T_{\text{ingest}}[\text{wiki}] > T_{\text{ingest}}[\text{rag}]$)	True (registered, and both weightings)
H3b supported ($T_{\text{query}}[\text{rag}] > T_{\text{query}}[\text{wiki}]$)	False ($21\times$ opposite direction)

H_3 as a conjunctive hypothesis requires both H_{3a} and H_{3b} to be supported, so H_{3b} ’s opposite-direction result alone is enough to refute it. The wiki cost intuition was that pre-compilation pays a large up-front fee in exchange for cheaper queries, breaking even at some N . The opposite happened: the user pays *more* per query, not less, so the preregistered wiki-amortization mechanism is impossible under the observed per-query costs. The preregistered crossover-queries formula’s denominator ($T_{\text{query}}[\text{rag}] - T_{\text{query}}[\text{wiki}]/13 = -121,020$) is negative; the formula returns a negative N , mathematically encoding “no positive-query crossover”.

H3a on the registered metric. The preregistered accounting sums prompt and completion tokens, and on that metric H_{3a} is adjudicated by the deposited artifacts with no reconstruction: $T_{\text{ingest}}[\text{wiki}] = 904,884,660$ against $T_{\text{ingest}}[\text{rag}] = 1,401,996$. **H3a is supported**, by a factor of 645. Nothing about prompt caching changes how many tokens the artifact records; caching changes what those tokens cost.

Post-hoc economic sensitivity. Because the wiki figure is cache-heavy, a raw token count overstates its economic weight relative to RAG’s uncached ingest, and a reader may reasonably ask whether the registered verdict is an artefact of that. It is not. The wiki artifact’s `tokens_in` aggregates uncached input, cache writes, and cache reads, which bill at $1\times$, $1.25\times$, and roughly $0.1\times$ of base input rate; the per-session split recovered from the build record shows cache reads dominate at 864.2M of 898.6M, or 96.2%. Re-weighting the input side gives billable-input-equivalents of 129.4M at the 5-minute cache-write rate and 155.2M at the 1-hour rate (the time-to-live in effect was not recorded, so we report the range rather than assume a value), and $T_{\text{ingest}}[\text{wiki}]$ of 135.7M to 161.5M once the 6.29M output tokens are added. **Output tokens are carried at face value, not weighted**, so these are not pure billable-cost figures either; output is 4.6% of the corrected total and weighting it changes no conclusion. H_{3a} holds on every reading, at a factor of 97 to 115 after weighting against 645 before it. Magnitudes are descriptive; the registered rule is directional. The H_3 verdict does not depend on any of this: the crossover denominator is the per-query differential, which is negative, so no positive ingest cost produces a positive crossover. H_3 remains refuted on H_{3b} , which is conjunctively sufficient.

5.2 Robustness checks (preregistered)

Only H_1 `inter_paper_mapping` (judge-avg) clears the preregistered “strongly corroborated” bar of $P(\mu \geq \theta) \geq 0.95$. The Bayesian posterior shrinks point estimates toward zero (e.g., H_1 primary `structural_integrity` sample mean $+2.000$ shrinks to posterior $+1.819$), reflecting the regularising prior at small n . The two robustness checks agree on direction but disagree on *how often* the threshold is robustly cleared.

Table 2: Bootstrap percentile CI (10,000 resamples) and preregistered Bayesian posterior ($\mu \sim \mathcal{N}(0, 4)$, $\sigma \sim \text{HalfNormal}(4)$, NUTS) for each H_1/H_2 cell. “Strong” marks $P(\mu \geq \theta) \geq 0.95$ per the preregistered Bayesian threshold ($H_1 \theta = +2$; $H_2 \theta = 0$).

Test	Reading	mean	bootstrap 95% CI	posterior μ	$P(\mu \geq \theta)$	strong?
H_1 struct	judge-avg	+1.625	[+0.75, +2.88]	+1.500	0.283	no
H_1 mapping	judge-avg	+6.625	[+4.13, +8.50]	+5.589	0.967	yes
H_1 struct	primary	+2.000	[+1.00, +3.25]	+1.819	0.440	no
H_1 mapping	primary	+6.250	[+3.50, +8.50]	+5.060	0.938	close
H_2 ground	judge-avg	+0.667	[−1.50, +2.00]	+0.522	0.660	no
H_2 ground	primary	+2.000	[+1.00, +3.00]	+1.808	0.937	close

5.3 Inter-rater reliability: a ceiling-judge finding

The preregistered IRR rule fired on all four criteria; max-deltas across the expanded $n = 13$ secondary coverage are 6 (**groundedness**), 4 (**structural_integrity**), 8 (**conflict_awareness**), 6 (**inter_paper_mapping**). Computed instead on the preregistered $n = 3$ IRR subset, max-deltas are 6 / 4 / 4 / 3, still triggering on all four criteria.

Max-delta is what the registered adjustment rule keys on, but it is a worst-case statistic that one cell can move and it says nothing about whether the judges rank systems alike. Table 3 therefore reports conventional agreement summaries over all 26 scored cells per criterion (13 questions $\times$ 2 systems), computed from the deposited judgments with no additional model calls.

Table 3: Inter-judge agreement per criterion, $n = 26$ cells (13 questions $\times$ 2 systems). ρ is Spearman, τ_b Kendall, QWK quadratic-weighted κ on the 1–10 scale, ICC(2,1) two-way random effects with absolute agreement. “Level” is $\text{mean}(\text{Gemini 2.5 Pro}) - \text{mean}(\text{GPT-5.4})$. “ ρ_Δ ” correlates the two judges’ *within-question* wiki–RAG differences ($n = 13$), i.e. agreement after any constant level shift is removed.

Criterion	ρ	τ_b	QWK	ICC	MAD	Level	ρ_Δ
groundedness	0.035	0.020	0.039	0.041	2.96	+2.27	0.323
conflict_awareness	0.406	0.275	0.239	0.247	2.04	+1.50	0.302
structural_integrity	0.653	0.471	0.443	0.452	1.35	+1.12	0.312
inter_paper_mapping	0.809	0.610	0.806	0.812	1.27	+0.19	0.705

Inter-judge agreement varied sharply by criterion, and the ordering is the same on every statistic: highest on the most operationally concrete criterion and lowest on the most holistic one. **inter_paper_mapping**, the one criterion with a countable definition (synthesis across ≥ 2 papers), reaches $\rho = 0.809$ and $\text{ICC} = 0.812$ with a level shift of only +0.19: the two judges substantially agree about both the level and the ordering. **groundedness**, the most holistic criterion, reaches $\rho = 0.035$ and $\text{ICC} = 0.041$ against a level shift of +2.27.

This corrects a weaker claim we made previously. An earlier version of this analysis, reasoning from max-deltas and criterion-level means, described the judges’ disagreement as “largely about level, not ranking.” That is true of **inter_paper_mapping** and defensible for **structural_integrity** ($\rho = 0.653$), but it is not true of **groundedness**, where the rank correlation is indistinguishable from zero and removing the level shift does not rescue it ($\rho_\Delta = 0.323$). On the criterion that carries H_2 , the two judges are not scoring a shared underlying quantity at different heights: they show essentially no rank agreement on this sample. The direction-flip already reported at the criterion level (GPT-5.4 rates RAG higher, 7.00 vs 6.31; Gemini 2.5 Pro rates wiki higher, 8.15

vs 9.69) is the visible symptom of that, and the H₂ subset shows it sharply: primary $\Delta = +2.000$, secondary $\Delta = -0.667$, judge-avg $\Delta = +0.667$, supported under the registered rule though the secondary judge alone would refute.

The per-judge per-system means (supplementary material) show the mechanism: Gemini 2.5 Pro ratings on the wiki output saturate near 10/10 across three of four criteria, while GPT-5.4 spreads between 6 and 9 with greater variance. `inter_paper_mapping` is again the exception, with the two judges’ RAG means differing by 0.15 (5.38 vs 5.23) and their wiki means by 0.54 (9.46 vs 10.00). We surface as a transferable methodological observation: *agreement was highest on the criterion with a concrete operational definition, in both level and rank, and lowest on the most holistic criterion, where it approached zero*, consistent with prior LLM-as-judge literature [Zheng et al., 2023, Liu et al., 2023b, Wang et al., 2024]. This suggests that criterion design, not judge identity alone, can materially affect LLM-judge reliability. We cannot isolate concreteness as the cause: there are only four criteria, concreteness was not manipulated, and score distribution, ceiling behaviour, task difficulty, and how each system expresses the property all co-vary with it. Establishing the causal claim would need the same criterion rewritten at several levels of operational specificity. The practical consequence does not depend on the mechanism: *criterion-level agreement should be measured rather than assuming a judge is uniformly reliable across a rubric*, and it is cheap to measure.

5.4 Post-hoc claim-level grounding

We ran a post-hoc claim-level grounding analysis on all 466 atomic claims. *This analysis is not preregistered* and is reported as exploratory robustness on the H₂ mechanism.

Important interpretive caveat. For RAG claims, the cited evidence is the original PDF chunk that the retrieval system surfaced; for wiki claims, the cited evidence is a wiki-page excerpt, which is itself a compilation of source PDFs and not an original PDF passage. The analysis below therefore tests *evidence-artifact claim alignment* (does each claim follow from what the system cited?), *not original-source fidelity* (does each claim follow from the underlying PDF?). A reader who wants source fidelity for wiki must additionally ask: does the wiki page itself accurately reflect the source PDF? We do not measure this; Section 7 flags it as the primary follow-up.

Aggregate by system. Total claims: RAG 150, Wiki 316. Cited rate (claims with non-empty `cited_source_idx`): RAG 88.0%, Wiki 76.3%. Wiki produces about twice as many claims per answer (24.3 vs 11.5) and cites a smaller fraction of them.

Answer length is an uncontrolled confound. Length here means the answer the judges read, measured on the answer text rather than on token telemetry. Per answer, single-round RAG writes 385 words (2,638 characters), decomp-RAG 604 (4,185), and the wiki 740 (5,250), so the wiki’s answers are 1.92× single-round RAG’s by word count and 1.99× by character count. Claims per answer run 11.5 / 17.8 / 24.3 and cited claims per answer 10.2 / 17.2 / 18.5 across the same three arms. Pipeline generation tokens (1,140 / 5,143 / 7,472) are a much larger spread and are *not* a length measure: they count every call a system makes, including the wiki’s browsing turns, decomp-RAG’s per-sub-question generations, and thinking tokens, none of which a judge sees.

Length is not controlled anywhere in this study and plausibly inflates the two criteria the wiki wins on, since a longer answer has more room to supply structure and cross-paper links, and more claims give more chances for a cited page to contain a supporting sentence. Two things bound how much it can explain. The claim-level rates are proportions, so length inflates numerator and

denominator alike, and the macro-averaged-by-question figures below weight all 13 answers equally and reach the same qualitative conclusion.

What we will not claim is that the ablation rules length out. Answer length increases monotonically across the three arms in the same order as synthesis performance, and decomp-RAG reaches 82% of the wiki’s word count while closing roughly 94% of the synthesis gap, which is as consistent with a length effect as against one. We treat length, and the prose-vs-bullet style difference it travels with, as an uncontrolled confound; a length- or retrieval-budget-matched replication is the cheapest experiment that would separate them.

Treating uncited claims as **unsupported** (*all-claims* view), the headline rates are: RAG 16.7% supported / 42.0% unsupported; Wiki 30.7% / 28.5%. The conditional-on-cited view (*citation-precision* sense) is:

Table 4: Claim-level verdicts, conditional on the claim carrying a citation. Claims were atomized by Claude Opus 4.7 and each scored by GPT-5.4 against the evidence its own answer cited; this axis is single-scorer, unlike the rubric scoring, which carries two-judge IRR coverage.

Verdict (conditional on cited)	RAG ($n = 132$)	Wiki ($n = 241$)
supported	18.9% (macro 18.4%)	40.2% (macro 37.3%)
partial	44.7%	53.1%
contradicted	2.3%	0.4%
unsupported	34.1% (macro 35.8%)	6.2% (macro 8.0%)

Macro-averages by question (each of 13 answers weighted equally regardless of claim count) are reported alongside the micro-averages to address the concern that long wiki answers might inflate wiki’s per-claim rate. The qualitative conclusion is invariant: wiki cited claims are *scored* supported $\sim 2\times$ more often, and scored unsupported $\sim 4\text{--}5\times$ less often, than RAG cited claims.

Of 4 directly-contradicted claims in the experiment, 3 are RAG (B1: HR/CI mismatch between cited table and answer; T4-cross-domain: feature-drift direction inverted; T5-glp1: false description of paper content) and 1 is Wiki (T2-oae: “in 95%” rendered as “over 95%”, a single-word paraphrase artefact).

By tier. Full rates by tier are in the supplementary material. The largest gap is on the *bias-check* tier, H_2 ’s home turf, where the two metrics point in opposite directions on the same tier: rubric $\Delta_{\text{groundedness}} = -2.00$ (RAG ahead, the registered metric) against claim-level supported% of 5.3% RAG vs 51.9% Wiki, and unsupported% of 78.9% RAG vs 7.4% Wiki. On multi-hop the rubric delta is larger still at -3.50 while the claim-level rates are far closer (32.0% vs 35.6%), so the divergence is not a constant offset but tracks how point-source the tier is.

Absolute-rate caveat. These rates should not be read as high absolute citation reliability: even the wiki’s cited claims are more often **partial** than strictly **supported** (53.1% vs 40.2% across all 13 questions). The result is a relative architecture comparison under a strict supported/partial boundary, not a benchmark-level claim of high citation precision.

Reading. The two metrics measure related but distinct properties. The rubric judge evaluates *holistic* answer-evidence trace and rewards RAG’s short, citation-heavy answers anchored to verbatim chunk text. The claim-level analysis evaluates *strict per-citation matching* and catches RAG’s tendency to retrieve a chunk and then synthesize/extrapolate beyond it, including the B1 confidence-interval error, where the chunk reports HR=1.37 but the answer says HR=1.38. The observed pattern is consistent with wiki compilation pre-positioning evidence in claim-shaped artifacts (so

when the answer cites a wiki page, the cited page typically contains the supporting sentence directly), although we do not measure header/claim alignment as an independent variable; this is not the same as source fidelity.

This does not invalidate the preregistered H_2 verdict. H_2 was a registered test on rubric groundedness and that verdict ($RAG \geq Wiki$) holds on the registered metric. It does materially nuance the underlying *mechanism*: the prereg conjectured that wiki’s lossy summarization would lose point-source fidelity; under claim-level scrutiny, RAG’s chunk-retrieve-and-synthesize pipeline shows worse *evidence-artifact alignment* than wiki’s compilation-and-traverse pipeline does. Whether wiki’s compilation step preserves source fidelity at a higher rate than RAG’s retrieval step is a separate question that this analysis does not adjudicate.

The methodological lesson: *rubric-style holistic grounding scores and claim-level citation alignment can disagree by direction, not just magnitude*. RAG/Wiki evaluations should report both.

5.5 Other preregistered exploratory analyses

The preregistration listed seven secondary analyses planned but not adjudicative. We summarize here; full data in the OSF supplement.

- *Bias-tag stratification*. The question taxonomy holds: bias=rag questions show RAG winning groundedness by 2.0; bias=wiki questions show Wiki winning all four criteria (largest on `inter_paper_mapping` at +5.75).
- *Latency*. RAG total 3.3 min, Wiki total 22.0 min ($6.6\times$); the wall-clock ratio is sublinear in tokens because adaptive thinking overlaps with input processing.
- *Adaptive-thinking engagement*. RAG triggered nonzero `thinking_tokens` on 2/13 questions; Wiki on 13/13. Both at the same `xhigh` flag: task shape, not flag, drives engagement.
- *Cost-vs-quality scatter*. RAG cluster ($\leq 8k$ tokens, scores 3.75–8.25) and Wiki cluster ($\sim 30k$ –210k tokens, 6.75–9.25) are not globally dominated. On 4 of 13 questions RAG matches or beats Wiki on rubric mean at an order-of-magnitude lower cost (T1-cardio, T3-rwd, B2, B3). The Pareto frontier is task-dependent.

6 Exploratory decomp-RAG ablation

A natural alternative hypothesis is that single-round retrieval is a known weakness on multi-hop synthesis [Trivedi et al., 2023, Jiang et al., 2023], so wiki’s H_1 advantage might be partly an artefact of the registered RAG configuration. We test this with a post-hoc decomposition-retrieval ablation.

Method. For each question: Claude Opus 4.7 decomposes it into 2–5 sub-questions; the existing single-round RAG pipeline (Section 3) runs per sub-question, keeping cited sources only; cited chunks are deduplicated; Claude Opus 4.7 generates the final answer using the *same* answer-generation system prompt as the single-round baseline (verbatim from the codebase). Same query model, corpus, vector store, and reranker; only the question-to-retrieval-context step changes. Three predictions were committed before running: *P1* decomp closes $\geq 50\%$ of the wiki-single-RAG gap on `inter_paper_mapping`; *P2* per-query token cost between 5–10 $\times$ single-round; *P3* claim-level supported-rate improves on multi-hop. Both judges were re-invoked on the (decomp, wiki) pair with the same prompts, rubric, and seeds as the main run: GPT-5.4 primary at `seed=42`, Gemini 2.5 Pro secondary at `seed=43`, medium reasoning effort, full 13-question coverage. **The preregistered**

max-delta IRR trigger fires on all four criteria here too (max-deltas 6 / 4 / 4 / 4 on groundedness / structural_integrity / conflict_awareness / inter_paper_mapping), so we report the ablation under the same **primary-only** and **IRR-adjusted** readings used in Section 5, with the judge-average as the headline. Applying the registered adjustment rule keeps the ablation on the same measurement scale as the confirmatory analysis; it does not make it a preregistered re-test, and *the results below remain exploratory and judge-sensitive*.

Table 5: Three-way comparison summary. Cost rows sum across the 13 questions. Synthesis-shape rows use the H_1 confirmatory subset ($n = 4$) and are reported under both readings: *primary* is GPT-5.4 alone, *judge-avg* is the IRR-adjusted mean of the two judges. Each gap is computed *within* a single judge run, so the Wiki column gives the wiki mean from the run the adjacent RAG arm was scored in; the two runs score identical wiki answers and agree exactly except on **structural_integrity** under judge-avg, where both values are shown as original / decomp. Claim-level row uses cited claims across all 13 questions and is single-scorer on every arm. ^aSingle-RAG, scored in the original judge run. ^bDecomp-RAG, scored in the post-hoc decomp judge run with the same prompts, rubric, and seeds.

Metric	Reading	Single ^a	Decomp ^b	Wiki	Wiki gap, single → decomp
Total query tokens	—	78k	491k	1,651k	decomp 3.4× cheaper than wiki on LLM tokens
Total wall-clock (min)	—	3.3	26.8	22.0	decomp slower than wiki (sequential)
H_1 inter_paper_mapping	primary	3.50	9.00	9.75	gap +6.25 → +0.75 (88% closed)
	judge-avg	3.25	9.50	9.88	gap +6.62 → +0.38 (94% closed)
H_1 structural_integrity	primary	7.00	8.75	9.00	gap +2.00 → +0.25 (88% closed)
	judge-avg	7.88	9.38	9.50 / 9.12	gap +1.62 → −0.25 (decomp edges above wiki)
Cited claim supported %	single-scorer	18.9	19.2	40.2	decomp does <i>not</i> close citation gap

Three-way result. Cost (P2). Decomp’s per-query LLM-token cost (under the H_3 token-accounting scope, excluding reranker and embedding spend) is 6.3× single-round and 3.4× cheaper than wiki, satisfying P2. Wall-clock is *slower* than wiki despite fewer tokens, because decomp performs sequential retrieval-then-rerank-then-validate calls per sub-question; the LLM-token picture is favorable for decomp, the wall-clock picture is not, and the embedding/reranker cost picture is not measured in this study.

Synthesis structure (P1). On the H_1 confirmatory subset, the decomp run supports P1 under both readings, and more strongly under the headline one. On **inter_paper_mapping** the wiki advantage shrinks from +6.62 to +0.38 judge-averaged, about 94% of the gap closed, and from +6.25 to +0.75 primary-only, about 88%. On **structural_integrity** it shrinks from +1.62 to −0.25 judge-averaged, the sign indicating that decomp edges slightly *above* the wiki, and from +2.00 to +0.25 primary-only. Against the decomp baseline, both criteria fall below the preregistered

+2.0 threshold under either reading; the wiki would not clear what the registered H_1 thresholds required in this comparison. *This does not alter the preregistered H_1 verdict against single-round RAG. It indicates that the synthesis-structure advantage is sensitive to the RAG baseline.*

Holistic groundedness, and a second instance of the ceiling-judge finding. The two judges disagree by direction on holistic groundedness in the ablation, exactly as they do in the main run (Section 5.3). GPT-5.4 scores decomp above the wiki by +1.15 across all 13 questions, Gemini 2.5 Pro scores the wiki above decomp by -0.85 , and the judge-average is +0.15, which we read as parity rather than as a decomp advantage. The same pattern holds on the tiers: +1.75 judge-averaged on multi-hop against +3.00 primary-only, and +0.67 on the H_2 bias-check parity test against +2.33 primary-only, still satisfying parity but with a much smaller margin. groundedness carries the largest inter-judge disagreement in the ablation as well (mean $|\Delta| = 2.54$ against 0.92 for `inter_paper_mapping`), reproducing the main run’s finding on an independent set of answers: *holistic groundedness is the criterion least robust to judge calibration, and a single-judge reading of it can invert.*

Claim-level grounding (P3 partial). We re-ran the §5.4 atomize-and-score pipeline, which is single-scorer (GPT-5.4) on every arm, on the 232 decomp claims (94.4% cited, vs 88.0% single, 76.3% wiki). Decomp’s overall cited-claim supported-rate is 19.2%, essentially equal to single-round (18.9%), and roughly half wiki’s (40.2%). The decomp claim-level advantage over single-round is concentrated on bias-check ($5.3 \rightarrow 43.5\%$, fixing chunk-recombination errors like the B1 confidence-interval mistake) and modestly on multi-hop ($32.0 \rightarrow 37.0\%$). On chronological / conflict / emergence / policy, decomp is similar to or slightly lower than single-round.

Reading. The pattern is consistent with two separable mechanisms underlying the wiki advantage. **Retrieval coverage** is substantially mitigated by decomposition retrieval in this ablation: when RAG breaks each question into sub-questions, it recovers nearly all of the wiki’s advantage on cross-paper synthesis, about 94% of the `inter_paper_mapping` gap under the adjusted reading. **Representation alignment** is *not* closed by broader retrieval: the wiki still does better at citing pages that directly support the specific claim being made. One plausible explanation is that wiki pages pre-position evidence in claim-shaped artifacts, while RAG chunks are aligned to the query rather than to each downstream answer claim. We do not directly measure this mechanism. Future work combining iterative retrieval with claim-grounded post-hoc citation repair would test whether the alignment gap can be closed too.

Caveats summary. Two-judge with the registered IRR adjustment applied, but the rubric scoring inherits the calibration drift documented in Section 5.3, which is direction-flipping on holistic groundedness here as well; the claim-level pipeline remains single-scorer on all three arms. Cross-run wiki absolute scores drift by at most 0.62 on any criterion’s 13-question mean under either judge and at most 0.15 under the judge-average; on the two criteria used in the H_1 rows the drift is 0.00 on `inter_paper_mapping` under both judges and at most 0.75 on `structural_integrity` under Gemini 2.5 Pro, which is why Table 5 computes every gap within a single judge run. $n = 4$ on the H_1 subset. Because the ablation is exploratory and small- n , we report point estimates only and treat them as mechanism evidence rather than confirmatory inference; one decomposition prompt was fixed before the run. Reproducer: `experiments/run_decomp_rag.py` + `run_claim_grounding_decomp.py` + `analyze_decomp_dual_judge.py`.

7 Discussion and limitations

What the findings mean. The wiki advantage appears to decompose into two separable mechanisms. The **retrieval-coverage** portion of the wiki’s synthesis-shape advantage is substantially mitigated by decomposition retrieval in our ablation (Section 6). The **representation-alignment** portion (the claim-level citation precision criterion) is *not* closed by broader retrieval. The observed pattern is consistent with wiki compilation pre-positioning evidence in claim-shaped artifacts (so when the answer cites a wiki page, the cited page typically contains the supporting sentence directly), while RAG retrieval aligns chunks with the *query*, not with each specific claim the answer ends up making. We do not measure header/claim alignment as an independent variable; this is a mechanism hypothesis consistent with the data, not a demonstrated causal account.

Karpathy’s three intuitions are partially borne out. (i) Pre-compilation helps with synthesis (supported on the registered baseline; substantially mitigable with decomp). (ii) Pre-compilation may sacrifice point-source fidelity (supported on the rubric; reversed under claim-level scrutiny). The relevant comparison there is evidence-artifact alignment, not original-PDF fidelity, which we do not measure. (iii) Pre-compilation shifts cost from query to ingest with finite-N amortization (refuted: per-query goes *up*, not down).

When to pick which. Single-round RAG for cost-sensitive point-source retrieval. Decomp-RAG for multi-paper synthesis where structure matters and the budget can absorb $\sim 6\times$ single-round LLM-token cost (excluding the additional reranker/embedding spend from multiple sub-question retrievals); tested at one operating point in this study. Wiki for evidence-artifact claim-citation alignment, despite the $\sim 21\times$ per-query token premium, pending source-PDF validation. The defensible form of H_3 ’s per-query result is narrow: *the tool-browsing wiki configuration tested here cannot amortize its ingest cost through query-token savings, because its measured per-query token use exceeds the tested RAG baseline’s rather than falling below it*. It does not generalize to compiled-knowledge architectures in general. Zhang [2026] report the opposite direction on query-time context with a wiki served as retrieved pages, and Chan et al. [2024] and Huerta [2026] serve compiled knowledge from a cached context window, which prices traversal out of the query path entirely. Agentic traversal is the cost driver we measured, not compilation.

What the findings do not show. *No human evaluation* (the most consequential gap): all rubric scoring, claim-level scoring, and decomp judging are LLM-based; the IRR finding (Section 5.3) shows LLM-judge calibration drift is large and direction-flipping on holistic criteria. Human validation is the highest-priority future work; until then the claim-level analysis should be read as LLM-scored *evidence-artifact alignment*, not human-validated source fidelity. The §6 decomp ablation now carries two-judge coverage and the same registered IRR adjustment as the confirmatory analysis, but it should still be read as exploratory and judge-sensitive: its holistic **groundedness** comparison reverses direction between the two judges, and its claim-level component remains single-scorer. *Small n* ($n = 4$ for H_1 , $n = 3$ for H_2): standard errors are large and the decomp ablation reports point estimates only. *Single corpus and single query model*: 24 papers, frontier LLM; transfer to large-corpus or different model families unverified. *Style and length confound*: wiki produces continuous prose, RAG bullet-and-cite, and the wiki’s answers are $1.9\times$ longer in visible text (740 vs 385 words per answer) with 24.3 claims against 11.5; neither is controlled for, and both plausibly favour the criteria the wiki wins on (Section 5.4). *One decomposer prompt* fixed before the run, not iterated. *Cost telemetry under prompt caching*: the registered H_3a verdict rests on the deposited artifacts alone, but the deposited aggregate does not license a *billable-cost* reading (Section 4), a caveat its manifest records at deposit time. The per-session components reconcile exactly to

that aggregate, so the quantity is recoverable and we report it in Section 5; the component table and a reconstruction script ship in the supplementary material, and the script refuses to report a figure unless the components sum to the deposited `totals.tokens_in` to the token. Three residual limits: the cache time-to-live was not recorded, so the write rate is bounded rather than known and we report both endpoints; output tokens are carried unweighted, so the corrected T_{ingest} is not a pure billable-cost figure; and the artifact carries no per-paper attribution of ingest tokens. None affects the H_{3a} direction, which holds by roughly two orders of magnitude, or H_{3b}, where query-side caching was disabled on both sides. *Methods note for replicators:* cached-vs-uncached cost-comparison studies must capture four columns per LLM call (uncached / cache_creation / cache_read / output), not a rolled-up `tokens_in`.

Beyond this study. Future work should compare claim-level grounding against original PDF passages (testing source fidelity for wiki) rather than against the cited evidence artifact only; extend claim-level scoring to a second scorer, the one axis of this study still resting on a single model; test retrieval-during-generation methods [Trivedi et al., 2023, Jiang et al., 2023] that may close more of the wiki advantage; separate the answer-length confound with a length- or retrieval-budget-matched replication, which would also test whether decomp-RAG’s synthesis gains are a chunk-exposure effect rather than a decomposition effect; and test hybrid wiki+source-RAG architectures that could backstop wiki’s rare paraphrase artefacts (1 of 4 contradicted claims) with source-chunk verification.

8 Conclusion

The most defensible conclusion of this preregistered comparison is not that compiled wiki memory beats RAG, nor the reverse. Grounded research synthesis is not a single capability: a system can organize evidence well, cite evidence well for each specific claim, or run cheaply, and in this experiment no architecture did all three best. Single-round RAG minimizes measured query LLM-token use; decomp-RAG matches the wiki on synthesis-shape rubric criteria, within 0.4 points on both under the judge-averaged reading, at $\sim 3.4\times$ lower per-query LLM-token cost; wiki retains the strongest LLM-scored evidence-artifact claim-citation alignment, at high per-query cost and pending source-PDF validation. Future small-corpus RAG/Wiki evaluations should report synthesis structure, claim-citation alignment, and cost separately rather than collapsing them into a single grounded-synthesis verdict.

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